

Arsheya Raj

raj94arsheya@gmail.com | +91 72500 32288 | [linkedin.com/in/rajarsheya](https://www.linkedin.com/in/rajarsheya) | [rajarsheya.github.io](https://github.com/rajarsheya) | github.com/rajarsheya

SUMMARY

Software Engineer with experience building scalable cloud-native applications, mobile applications, distributed systems, and AI-powered solutions using C++, Python, React Native, Google Cloud Platform, and AWS.

EDUCATION

University of Washington – School of STEM (Bothell)	Bothell, WA
Master of Science in Computer Science and Software Engineering (GPA: 3.9/4.0)	August 2024
Relevant Coursework: Algorithm Design & Analysis, Mobile Computing, High Performance Computing (HPC), Evaluating Software Design, Software Management, Advanced Topics in Computer Vision.	
Birla Institute of Technology Mesra	Ranchi, India
Bachelor of Engineering in Information Technology (GPA: 3.82/4.0 (WES ICAP))	June 2018
Relevant Coursework: Data Mining Concepts and Techniques, Parallel & Distributed Computing, Principles of Soft Computing, Computer Graphics & Multimedia, Artificial Intelligence	

TECHNICAL SKILLS

<i>Programming Languages:</i>	C/C++, SQL/PostgreSQL, Python, JavaScript, Java, C#, Shell Scripting, Swift
<i>Frameworks:</i>	React Native, React.js, Flask, NumPy, Pandas
<i>Cloud Computing Platforms:</i>	Amazon Web Services (AWS), Google Cloud Platform (GCP), Firebase
<i>Technical Skills:</i>	Data Structures & Algorithms, System Design, NVIDIA CUDA, OpenCV, Unity, MATLAB, PyTorch, TensorFlow, Operating Systems, Object-Oriented Programming, REST APIs, Microservices, Git, Linux, Pub/Sub, Dataflow, BigQuery, Terraform
<i>Certifications:</i>	Google Cloud Certified - Professional Cloud Architect (Credential ID: a10a7b621dd94a6bb481b30fbd324b8c) [Feb 2026 - Feb 2028] Google Cloud Certified - Associate Cloud Engineer (Credential ID: f2e23bbe7a6041689d3eeb1464245028) [Feb 2026 - Feb 2029] AWS Certified Solution Architect Engineer - Associate (Credential ID: 4b9101bf86204cfb81f9ae1c161a5cee) [July 2025 - July 2028] Google Cloud Certified - Digital Leader Developer (Credential ID: bb87ca45c3a74d24a5f42851a2c1f6d6) [Nov 2022 - Sept2028] See LinkedIn for more certifications (linkedin.com/in/rajarsheya)
<i>Achievements:</i>	Qualified for ACM ICPC Onsite Round and Google Code Jam in 2016 and 2017 , Participated in SnackDown 2016, 2017, and 2021 conducted by Codechef. Qualified RMO (Regional Mathematical Olympiad) 2013 and participated in INMO (Indian National Mathematical Olympiad) 2014 .

EXPERIENCE

Co-founder (Nov 2024- Present) & Chief Technology Officer (Nov 2024 - Sept 2025) - Vaccine Genie, Community Family and Internal Medicine	
- Vaccine Genie is a secure, scalable vaccination record platform to improve healthcare data management, prevent redundancy, support disease prevention, and advance medical research for better outcomes. - Led development of a cloud-native vaccination record platform using React Native, Google Cloud Platform, Gemini AI, and FHIR/HL7 healthcare standards. - Designed and implemented scalable REST APIs and microservices supporting secure EHR interoperability and high-volume healthcare data exchange. - Architected cloud infrastructure supporting secure storage, retrieval, and synchronization of vaccination records. Achievements: Winner – Sacia Digital Health Challenge 2023, University of Washington. Participant – Hollomon Health Innovation Challenge 2024 and 2025, Attended CES 2025. Recognized as a resource by the Washington State Department of Health website.	
I-Corps Summer Cohort 2023 , CoMotion at University of Washington	June 2023 - July2023
Conducted 25+ customer discovery interviews and applied Lean Canvas to define target customers, early adopters, and essential features.	
Freelance App Developer / React Native Developer , Self-Employed	April 2020 - April 2022
Developed Android and iOS applications using React Native, JavaScript, and Google Cloud Platform, incorporating features like video conferencing, content sharing, and quiz portals.	

- Conducted thorough **alpha, beta, and closed testing phases** to ensure app quality, performance, and security.
- Contributed to the creation of 6 **system design specifications** for client web and native applications, streamlining development by ensuring technical feasibility and alignment with business needs.
- Diagnosed and fixed 10 critical bugs in a client's quiz-portal React Native app, enhancing functionality and user experience.

Business Technology Analyst, ZS Associates

May 2018 - September 2019

- **Developed** automated workflows and scalable backend solutions for risk-based pricing models, **helping a pharmaceutical client optimize profitability and decision-making.**
- **Collaborated** on three successful releases and enhanced backend development processes with AWS services (**PostgreSQL** in Amazon **Redshift**, **AWS RDS**, **EC2**, **S3**), **improving system scalability and performance.**
- **Automated ETL batch uploads** on AWS EC2 instances using Shell scripting, **reducing runtime by 5% and boosting operational efficiency.**
- **Helped integrate** data from multiple sources into the Reltio Master Data Management (MDM) system using Node.js, JSON, and Java (Reltio JARs), **creating a unified, user-friendly data interface.**

Business Technology Analyst Intern, ZS Associates

Winter 2018

- Designed and implemented scalable data solutions using **PostgreSQL, ETL workflows, and Amazon Redshift**, **optimizing data warehousing** and processing for the Biosimilars release.
- Collaborated with leadership, clients, and colleagues to present key insights, test case results, and performance analyses, driving informed decision-making and ensuring seamless deployment.

CSS Grader and Peer Facilitator (Teaching Assistant), University of Washington Bothell

- Database Systems (CSS 475) Autumn 2023
 - Taught **NoSQL databases** to 40+ students, facilitating their understanding of **CRUD operations, ACID principles, Relational Algebra, Normalization, CAP Theorem** and **ER diagrams.**
- Operating Systems (CSS 430 A) Winter 2024
 - Assisted 40+ students in understanding concepts like **Scheduling, Synchronization, Deadlock, Memory Management**, and **Paging.**
- High-Performance Computing (CSS535) Winter 2024
 - Supported 20+ students in learning of advanced HPC topics including **BLAS, GPU hardware models, Threads and Blocks, Kernels, CUDA programming models, Profiling (Nsight Compute and Nsight Systems)**, and **GPU patterns.**

Summer Intern: Big Data, AllSoft Solutions Pvt. Ltd.

Summer 2017

- **Developed proof-of-concept proposal for analyzing large YouTube's datasets** using **Hadoop, HDFS, Map-Reduce, Hive, Pig**, and other technologies to curate top 10 lists across various categories.

TECHNICAL PROJECTS AND MASTER'S THESIS

[Project] Cloud-Based Resume Intelligence Assistant

Dec 2025

- Built a GenAI-powered application using Python, open-source LLMs, and semantic similarity techniques to extract skills, identify gaps, and generate ATS-friendly resume recommendations.

[Project] Chatbot Response Acceleration with CUDA LLM Inference

May 2025

- Accelerated local LLM inference using NVIDIA CUDA, reducing response latency and improving throughput for real-time chatbot interactions.

[Project] Real-time Traffic Analysis with CUDA Object Detection

March 2025

- Developed a CUDA-accelerated computer vision pipeline using YOLO for real-time vehicle detection, counting, and speed estimation.

[Master's Thesis] Innovative Rehabilitation Approach for Upper Limb Neurologic Conditions Using Mixed-Reality Simulation and EEG/EMG Biofeedback

- Developed **AR/MR** rehabilitation environment for stroke patients as part of the Smart NeuroRehab Ecosystem.
- Built end-to-end biosignal processing pipelines for **EEG** and **EMG** data in **MATLAB**, training and evaluating **CNN, RNN, LSTM**, and **SVM** models in **PyTorch** and **TensorFlow** for motor-task classification.
- Evaluated multiple deep learning architectures for biosignal-based movement recognition.

[Project] AI Startup Intelligence Platform

June 2026

- Built an event-driven startup intelligence platform using **Pub/Sub, Dataflow, BigQuery**, and **GKE** to ingest, process, and analyze funding, hiring, and company-growth signals in real time, with **RAG**-based semantic search for trend discovery.